

Initiating Search

November 10, 2025, 2:35 PM

• Search:

CAS Lexicon Search:

Ley-Griffith oxidation - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (18)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (56)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%	
Document Type:	Journal	

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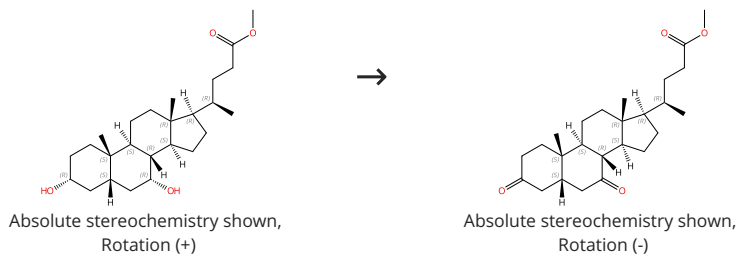
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Reactions (56)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (52)

 Suppliers (11)

31-614-CAS-34334478

Steps: 1 Yield: 100%

Selectivity of Oxidizing Agents toward Axial and Equatorial Hydroxyl Groups

By: Kaspar, Miroslav; et al

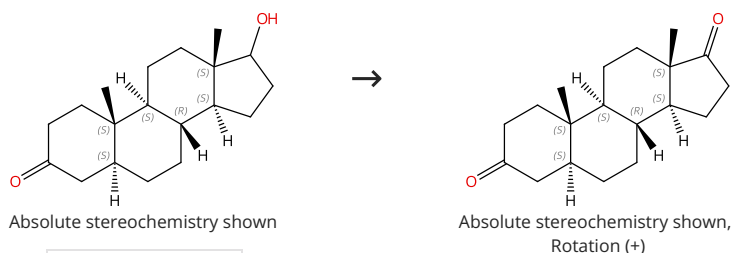
Journal of Organic Chemistry (2022), 87(14), 9157-9170.

- 1.1 Reagents: Chromium trioxide
Solvents: Acetone, Water; 0 °C; 30 min, 0 °C
- 1.2 Reagents: Isopropanol

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 99%


 Supplier (1)

 Suppliers (26)

31-480-CAS-18797020

Steps: 1 Yield: 99%

ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis

By: Moore, Peter W.; et al

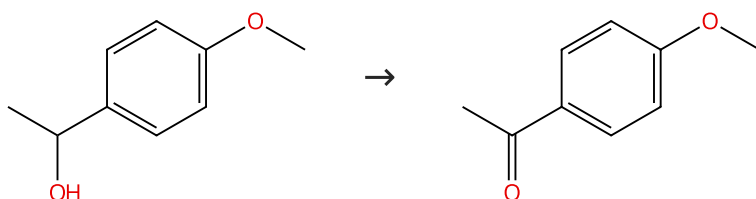
Chemistry - A European Journal (2018), 24(18), 4556-4561.

- 1.1 Reagents: N-methylmorpholine N-oxide
Solvents: Dichloromethane; 1 h, rt
- 1.2 Catalysts: TPAP; 1.5 h, rt

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 99%

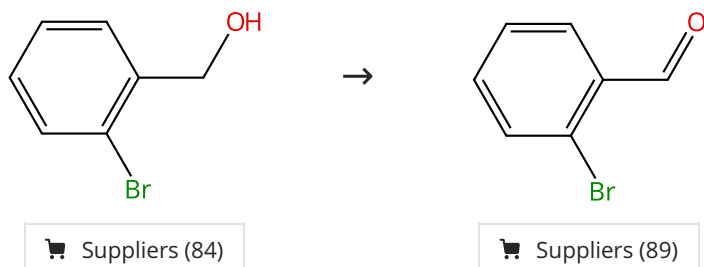

 Suppliers (66)

 Suppliers (98)

31-480-CAS-22342262	Steps: 1 Yield: 99%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: (OC-6-42)-[2,6-Bis[5-butyl-1-(2-propen-1-yl)-1 <i>H</i> -pyrazol-3-yl-κ <i>N</i> ²]pyridine-κ <i>M</i>]dichloro(triphenylphosphine) ruthenium Solvents: Heptane; 10 min, 80 °C		By: Weingart, Pascal; et al ChemCatChem (2020), 12(15), 3919-3928.

Scheme 4 (1 Reaction)

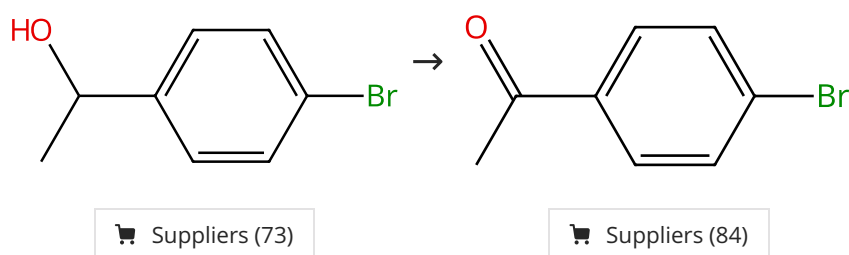
Steps: 1 Yield: 99%



31-480-CAS-22342271	Steps: 1 Yield: 99%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: (OC-6-42)-[2,6-Bis[5-butyl-1-(2-propen-1-yl)-1 <i>H</i> -pyrazol-3-yl-κ <i>N</i> ²]pyridine-κ <i>M</i>]dichloro(triphenylphosphine) ruthenium Solvents: Heptane; 10 min, 80 °C		By: Weingart, Pascal; et al ChemCatChem (2020), 12(15), 3919-3928.

Scheme 5 (1 Reaction)

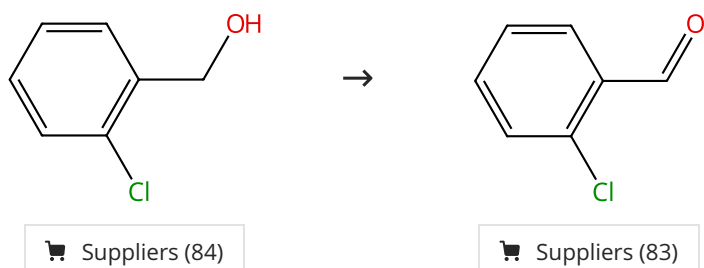
Steps: 1 Yield: 99%



31-480-CAS-22342259	Steps: 1 Yield: 99%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: Ruthenium trichloride hydrate Solvents: Dimethylformamide; 10 min, 60 °C		By: Weingart, Pascal; et al ChemCatChem (2020), 12(15), 3919-3928.

Scheme 6 (1 Reaction)

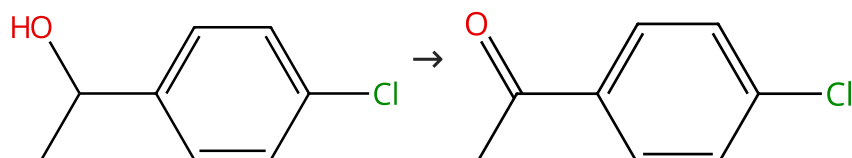
Steps: 1 Yield: 99%



31-480-CAS-22342270	Steps: 1 Yield: 99%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: (OC-6-42)-[2,6-Bis[5-butyl-1-(2-propen-1-yl)-1 <i>H</i> -pyrazol-3-yl-κ ^{N2}]pyridine-κ ^M]dichloro(triphenylphosphine) ruthenium Solvents: Heptane; 10 min, 80 °C		By: Weingart, Pascal; et al ChemCatChem (2020), 12(15), 3919-3928.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 99%



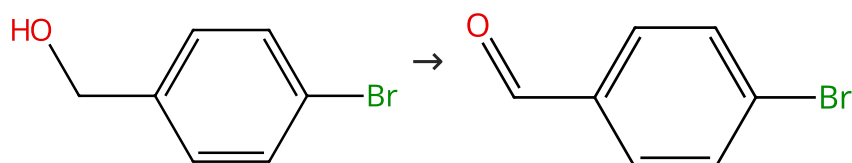
Suppliers (67)

Suppliers (67)

31-480-CAS-22342260	Steps: 1 Yield: 99%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: (OC-6-42)-[2,6-Bis[5-butyl-1-(2-propen-1-yl)-1 <i>H</i> -pyrazol-3-yl-κ ^{N2}]pyridine-κ ^M]dichloro(triphenylphosphine) ruthenium Solvents: Heptane; 10 min, 80 °C		By: Weingart, Pascal; et al ChemCatChem (2020), 12(15), 3919-3928.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 99%



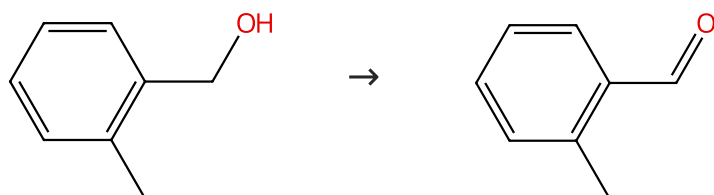
Suppliers (88)

Suppliers (91)

31-480-CAS-22342267	Steps: 1 Yield: 99%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: (OC-6-42)-[2,6-Bis[5-butyl-1-(2-propen-1-yl)-1 <i>H</i> -pyrazol-3-yl-κ ^{N2}]pyridine-κ ^M]dichloro(triphenylphosphine) ruthenium Solvents: Heptane; 10 min, 80 °C		By: Weingart, Pascal; et al ChemCatChem (2020), 12(15), 3919-3928.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 99%



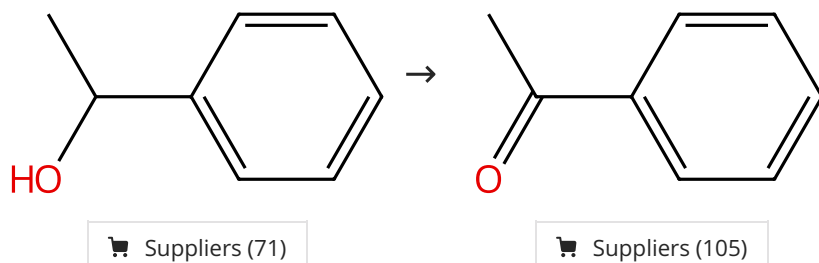
Suppliers (76)

Suppliers (87)

31-480-CAS-22342269	Steps: 1 Yield: 99%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: (OC-6-42)-[2,6-Bis[5-butyl-1-(2-propen-1-yl)-1 <i>H</i> -pyrazol-3-yl-κ <i>N</i> ²]pyridine-κ <i>M</i>]dichloro(triphenylphosphine) ruthenium Solvents: Heptane; 10 min, 80 °C		By: Weingart, Pascal; et al ChemCatChem (2020), 12(15), 3919-3928.

Scheme 10 (1 Reaction)

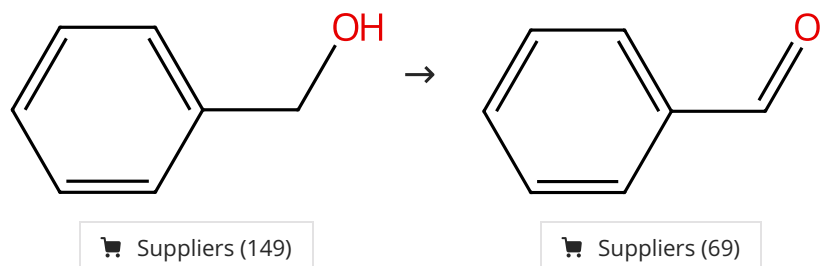
Steps: 1 Yield: 99%



31-480-CAS-22342258	Steps: 1 Yield: 99%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: (OC-6-42)-[2,6-Bis[5-butyl-1-(2-propen-1-yl)-1 <i>H</i> -pyrazol-3-yl-κ <i>N</i> ²]pyridine-κ <i>M</i>]dichloro(triphenylphosphine) ruthenium Solvents: Heptane; 10 min, 80 °C		By: Weingart, Pascal; et al ChemCatChem (2020), 12(15), 3919-3928.

Scheme 11 (1 Reaction)

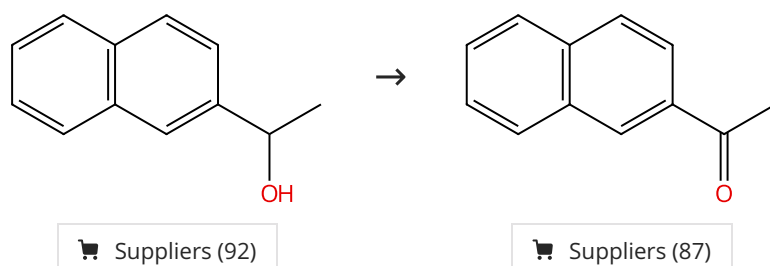
Steps: 1 Yield: 99%



31-480-CAS-22342266	Steps: 1 Yield: 99%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: (OC-6-42)-[2,6-Bis[5-butyl-1-(2-propen-1-yl)-1 <i>H</i> -pyrazol-3-yl-κ <i>N</i> ²]pyridine-κ <i>M</i>]dichloro(triphenylphosphine) ruthenium Solvents: Heptane; 10 min, 80 °C		By: Weingart, Pascal; et al ChemCatChem (2020), 12(15), 3919-3928.

Scheme 12 (1 Reaction)

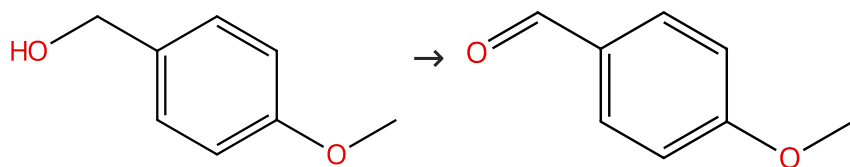
Steps: 1 Yield: 99%



31-480-CAS-22342265	Steps: 1 Yield: 99%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: Ruthenium trichloride hydrate Solvents: Dimethylformamide; 10 min, 60 °C		By: Weingart, Pascal; et al ChemCatChem (2020), 12(15), 3919-3928.

Scheme 13 (2 Reactions)

Steps: 1 Yield: 88-99%



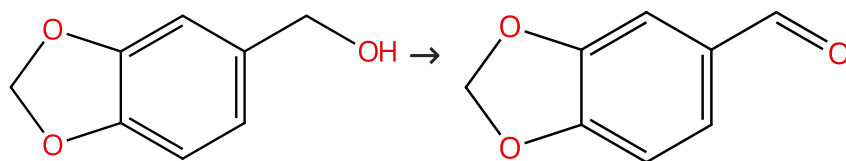
Suppliers (105)

Suppliers (128)

31-480-CAS-22342268	Steps: 1 Yield: 99%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: (OC-6-42)-[2,6-Bis[5-butyl-1-(2-propen-1-yl)-1 <i>H</i> -pyrazol-3-yl-κ <i>N</i> ²]pyridine-κ <i>N</i>]dichloro(triphenylphosphine) ruthenium Solvents: Heptane; 10 min, 80 °C		By: Weingart, Pascal; et al ChemCatChem (2020), 12(15), 3919-3928.
31-480-CAS-15777410	Steps: 1 Yield: 88%	Selectivity Modulation of the Ley-Griffith TPAP Oxidation with N-Oxide Salts
1.1 Reagents: 4-Aza-1-azoniabicyclo[2.2.2]octane, 1-hydroxy-, tetraphenylborate(1-), compd. with 1,4-diazabicyclo[2.2.2]octane 1-oxide (1:1:1) Solvents: Acetonitrile; 10 min, rt		By: Moore, Peter W.; et al European Journal of Organic Chemistry (2016), 2016(20), 3401-3407.
1.2 Catalysts: TPAP; 16 h, rt		
1.3 Solvents: Water; rt		
Experimental Protocols		

Scheme 14 (13 Reactions)

Steps: 1 Yield: 88-97%



Suppliers (80)

Suppliers (42)

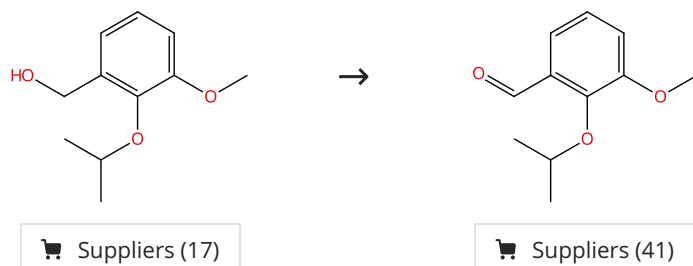
31-480-CAS-15777411	Steps: 1 Yield: 97%	Selectivity Modulation of the Ley-Griffith TPAP Oxidation with N-Oxide Salts
1.1 Reagents: 4-Aza-1-azoniabicyclo[2.2.2]octane, 1-hydroxy-, tetraphenylborate(1-), compd. with 1,4-diazabicyclo[2.2.2]octane 1-oxide (1:1:1) Solvents: Acetonitrile; 10 min, rt		By: Moore, Peter W.; et al European Journal of Organic Chemistry (2016), 2016(20), 3401-3407.
1.2 Catalysts: TPAP; 16 h, rt		
1.3 Solvents: Water; rt		
Experimental Protocols		

31-480-CAS-18797033	Steps: 1 Yield: 95%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: 2204287-75-0; 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		
31-480-CAS-18797008	Steps: 1 Yield: 95%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: 2204287-69-2; 18.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		
31-480-CAS-18797027	Steps: 1 Yield: 92%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: Phosphonium, tetraphenyl-, (<i>T</i> -4)-tetraoxoruthenate(1-) (1:1); 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		
31-480-CAS-18797032	Steps: 1 Yield: 92%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: 2204287-74-9; 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		
31-480-CAS-18797029	Steps: 1 Yield: 92%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: Phosphorus(1+), triphenyl(<i>P,P,P</i> -triphenylphosphine imidato- <i>N</i>)-, (<i>T</i> -4)-, (<i>T</i> -4)-tetraoxoruthenate(1-); 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		
31-480-CAS-18797028	Steps: 1 Yield: 91%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: Arsonium, tetraphenyl-, (<i>T</i> -4)-tetraoxoruthenate(1-); 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		
31-480-CAS-18797034	Steps: 1 Yield: 91%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: 2204287-76-1; 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		
31-480-CAS-18797026	Steps: 1 Yield: 90%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: 1-Butanaminium, <i>N,N,N</i> -tributyl-, (<i>T</i> -4)-tetraoxoruthenate(1-) (1:1); 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		

31-480-CAS-18797036	Steps: 1 Yield: 90%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: 2204287-78-3; 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		
31-480-CAS-18797031	Steps: 1 Yield: 89%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: 2204287-71-6; 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		
31-480-CAS-18797035	Steps: 1 Yield: 88%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: 2204287-77-2; 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		
31-480-CAS-18797030	Steps: 1 Yield: 88%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: 2204287-70-5; 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		

Scheme 15 (2 Reactions)

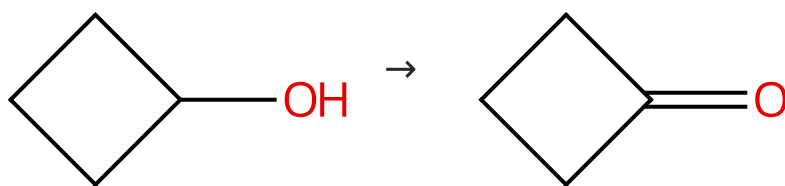
Steps: 1 Yield: 89-95%



31-480-CAS-18797009	Steps: 1 Yield: 95%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: 1960452-14-5 Solvents: Acetonitrile; 10 min, rt		By: Moore, Peter W.; et al
1.2 Catalysts: TPAP; 16 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
1.3 Reagents: Water; rt		
Experimental Protocols		
31-480-CAS-15777393	Steps: 1 Yield: 89%	Selectivity Modulation of the Ley-Griffith TPAP Oxidation with N-Oxide Salts
1.1 Reagents: 4-Aza-1-azoniabicyclo[2.2.2]octane, 1-hydroxy-, tetraphenylborate(1-), compd. with 1,4-diazabicyclo[2.2.2]octane 1-oxide (1:1:1) Solvents: Acetonitrile; 10 min, rt		By: Moore, Peter W.; et al
1.2 Catalysts: TPAP; 16 h, rt		European Journal of Organic Chemistry (2016), 2016(20), 3401-3407.
1.3 Solvents: Water; rt		
Experimental Protocols		

Scheme 16 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (82)

Suppliers (81)

31-480-CAS-19597879

Steps: 1 Yield: 95%

Hydrogen-Bonding Interactions in the Ley-Griffith Oxidation: Practical Considerations for the Synthetic Chemist

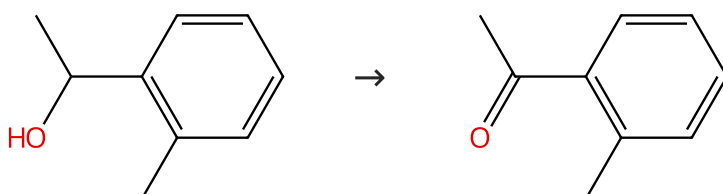
By: Moore, Peter W.; et al

European Journal of Organic Chemistry (2019), 2019(2-3), 303-308.

1.1 **Reagents:** N-methylmorpholine N-oxide
Catalysts: TPAP
Solvents: Acetonitrile; 1 - 8 h, rt

Scheme 17 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (64)

Suppliers (86)

31-480-CAS-22342263

Steps: 1 Yield: 95%

Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols

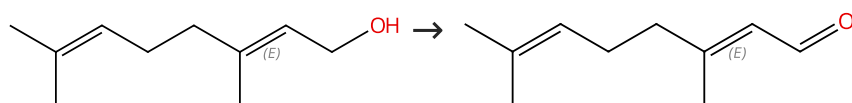
By: Weingart, Pascal; et al

ChemCatChem (2020), 12(15), 3919-3928.

1.1 **Reagents:** Trimethylamine oxide
Catalysts: Ruthenium trichloride hydrate
Solvents: Dimethylformamide; 10 min, 60 °C

Scheme 18 (1 Reaction)

Steps: 1 Yield: 95%



Double bond geometry shown

Suppliers (117)

Double bond geometry shown

Suppliers (36)

31-480-CAS-18797014

Steps: 1 Yield: 95%

ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis

By: Moore, Peter W.; et al

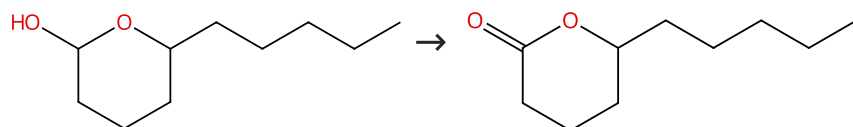
Chemistry - A European Journal (2018), 24(18), 4556-4561.

1.1 **Reagents:** N-methylmorpholine N-oxide
Solvents: Dichloromethane; 1 h, rt
 1.2 **Catalysts:** TPAP; 0.5 h, rt

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (5)

Suppliers (78)

31-480-CAS-18797019

Steps: 1 Yield: 94%

ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis

By: Moore, Peter W.; et al

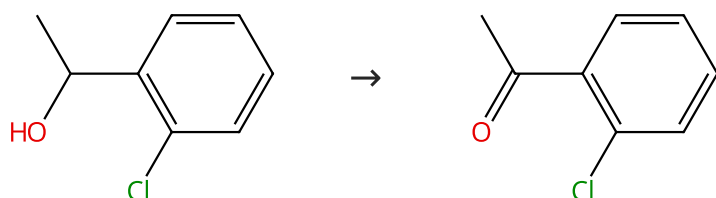
Chemistry - A European Journal (2018), 24(18), 4556-4561.

1.1 **Reagents:** N-methylmorpholine N-oxide**Solvents:** Dichloromethane; 1 h, rt1.2 **Catalysts:** 2204287-72-7; 0.5 h, rt

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (71)

Suppliers (85)

31-480-CAS-22342264

Steps: 1 Yield: 94%

Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols

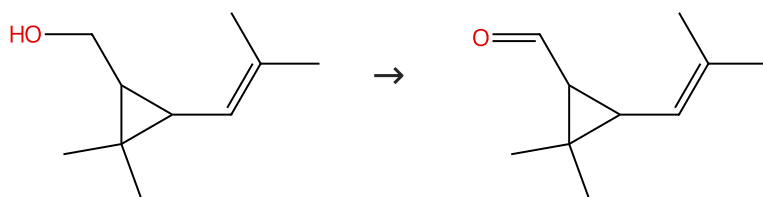
By: Weingart, Pascal; et al

ChemCatChem (2020), 12(15), 3919-3928.

1.1 **Reagents:** Trimethylamine oxide**Catalysts:** Ruthenium trichloride hydrate**Solvents:** Dimethylformamide; 10 min, 60 °C

Scheme 21 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (41)

Suppliers (4)

31-480-CAS-18797016

Steps: 1 Yield: 92%

ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis

By: Moore, Peter W.; et al

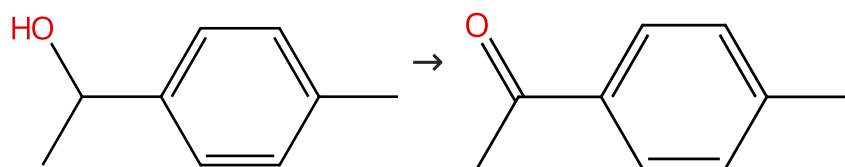
Chemistry - A European Journal (2018), 24(18), 4556-4561.

1.1 **Reagents:** N-methylmorpholine N-oxide**Solvents:** Dichloromethane; 1 h, rt1.2 **Catalysts:** 2204287-69-2; 0.5 h, rt

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (69)

Suppliers (103)

31-480-CAS-22342261

Steps: 1 Yield: 91%

Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols

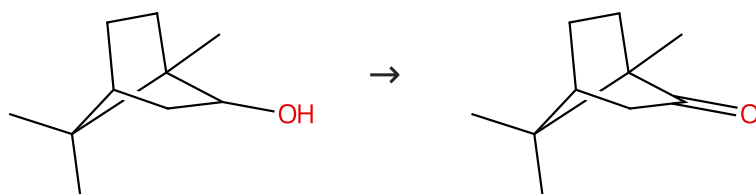
- 1.1 **Reagents:** Trimethylamine oxide
Catalysts: Ruthenium trichloride hydrate
Solvents: Dimethylformamide; 10 min, 60 °C

By: Weingart, Pascal; et al

ChemCatChem (2020), 12(15), 3919-3928.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (34)

Suppliers (127)

31-480-CAS-18797017

Steps: 1 Yield: 91%

ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis

- 1.1 **Reagents:** N-methylmorpholine N-oxide
Solvents: Dichloromethane; 1 h, rt

By: Moore, Peter W.; et al

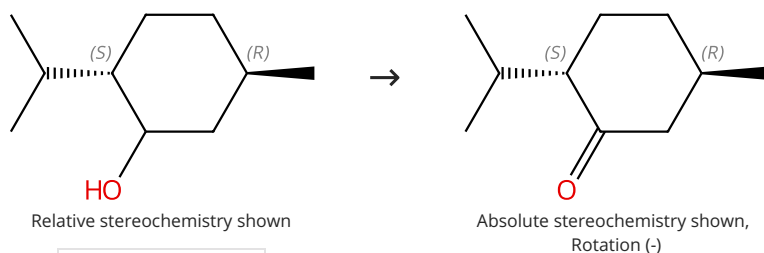
- 1.2 **Catalysts:** 2204287-69-2; 0.5 h, rt

Chemistry - A European Journal (2018), 24(18), 4556-4561.

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 90%



Relative stereochemistry shown

Absolute stereochemistry shown,
Rotation (-)

Suppliers (7)

Suppliers (76)

31-480-CAS-18797022

Steps: 1 Yield: 90%

ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis

- 1.1 **Solvents:** Dichloromethane; 1 h, rt

By: Moore, Peter W.; et al

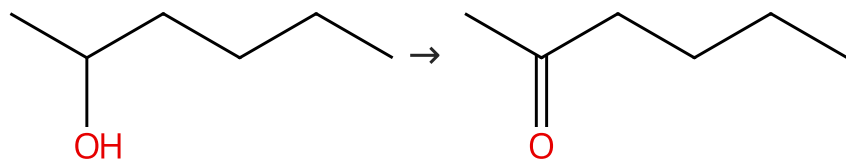
- 1.2 **Reagents:** Oxygen
Catalysts: TPAP; 3 h, rt

Chemistry - A European Journal (2018), 24(18), 4556-4561.

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 89%



Suppliers (53)

Suppliers (70)

31-480-CAS-22342273

Steps: 1 Yield: 89%

Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols

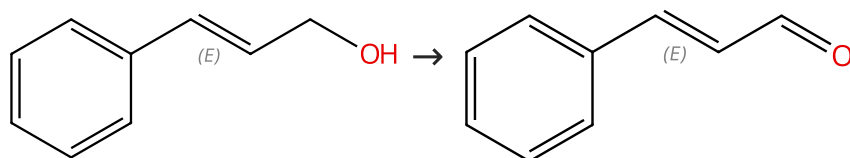
By: Weingart, Pascal; et al

ChemCatChem (2020), 12(15), 3919-3928.

- 1.1 **Reagents:** Trimethylamine oxide
Catalysts: Ruthenium trichloride hydrate
Solvents: Dimethylformamide; 10 min, 60 °C

Scheme 26 (2 Reactions)

Steps: 1 Yield: 82-89%



Double bond geometry shown

Suppliers (66)

Double bond geometry shown

Suppliers (109)

31-480-CAS-18797013

Steps: 1 Yield: 89%

ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis

By: Moore, Peter W.; et al

Chemistry - A European Journal (2018), 24(18), 4556-4561.

- 1.1 **Reagents:** N-methylmorpholine N-oxide
Solvents: Dichloromethane; 1 h, rt

- 1.2 **Catalysts:** 2204287-69-2; 0.5 h, rt

Experimental Protocols

31-480-CAS-15777414

Steps: 1 Yield: 82%

Selectivity Modulation of the Ley-Griffith TPAP Oxidation with N-Oxide Salts

By: Moore, Peter W.; et al

European Journal of Organic Chemistry (2016), 2016(20), 3401-3407.

- 1.1 **Reagents:** 4-Aza-1-azoniabicyclo[2.2.2]octane, 1-hydroxy-, tetraphenylborate(1-), compd. with 1,4-diazabicyclo[2.2.2]octane 1-oxide (1:1:1)
Solvents: Acetonitrile; 10 min, rt

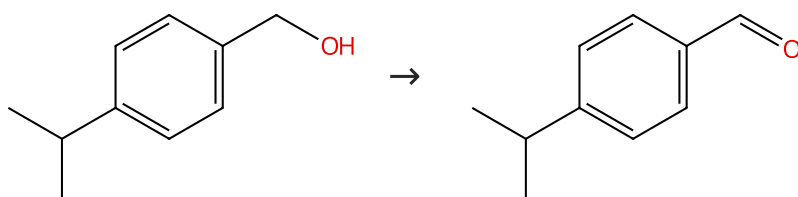
- 1.2 **Catalysts:** TPAP; 16 h, rt

- 1.3 **Solvents:** Water; rt

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 88%



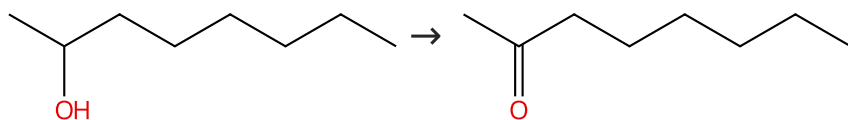
Suppliers (97)

Suppliers (92)

31-480-CAS-15777418	Steps: 1 Yield: 88%	Selectivity Modulation of the Ley-Griffith TPAP Oxidation with N-Oxide Salts
1.1 Reagents: 4-Aza-1-azoniabicyclo[2.2.2]octane, 1-hydroxy-, tetraphenylborate(1-), compd. with 1,4-diazabicyclo[2.2.2]octane 1-oxide (1:1:1) Solvents: Acetonitrile; 10 min, rt		By: Moore, Peter W.; et al
1.2 Catalysts: TPAP; 16 h, rt		European Journal of Organic Chemistry (2016), 2016(20), 3401-3407.
1.3 Solvents: Water; rt		
Experimental Protocols		

Scheme 28 (1 Reaction)

Steps: 1 Yield: 88%



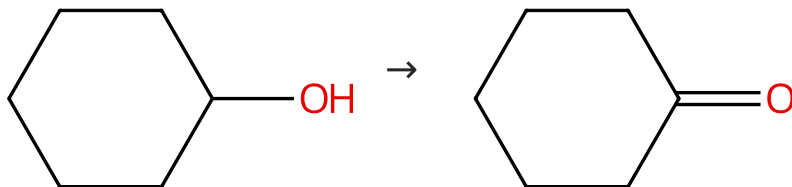
Suppliers (90)

Suppliers (87)

31-480-CAS-18797018	Steps: 1 Yield: 88%	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis
1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt		By: Moore, Peter W.; et al
1.2 Catalysts: 2204287-69-2; 0.5 h, rt		Chemistry - A European Journal (2018), 24(18), 4556-4561.
Experimental Protocols		

Scheme 29 (1 Reaction)

Steps: 1 Yield: 87%



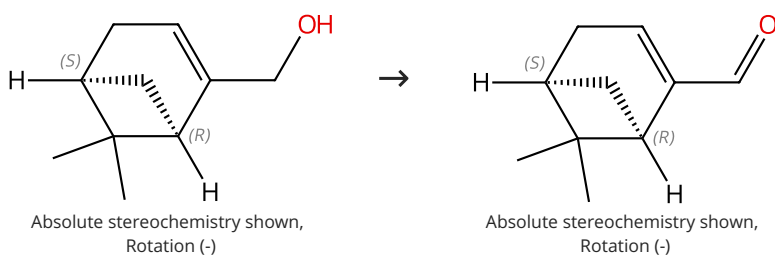
Suppliers (81)

Suppliers (102)

31-480-CAS-22342272	Steps: 1 Yield: 87%	Two Simple and Highly Efficient Variants of the Griffith-Ley Oxidation of Alcohols
1.1 Reagents: Trimethylamine oxide Catalysts: Ruthenium trichloride hydrate Solvents: Dimethylformamide; 10 min, 60 °C		By: Weingart, Pascal; et al
		ChemCatChem (2020), 12(15), 3919-3928.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 85%

Absolute stereochemistry shown,
Rotation (-)Absolute stereochemistry shown,
Rotation (-)

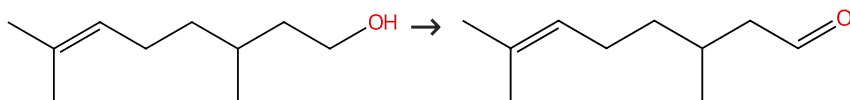
Suppliers (47)

Suppliers (53)

31-480-CAS-15777417 Steps: 1 Yield: 85% 1.1 Reagents: 4-Aza-1-azoniabicyclo[2.2.2]octane, 1-hydroxy-, tetraphenylborate(1-), compd. with 1,4-diazabicyclo[2.2.2]octane 1-oxide (1:1:1) Solvents: Acetonitrile; 10 min, rt 1.2 Catalysts: TPAP; 16 h, rt 1.3 Solvents: Water; rt Experimental Protocols	Selectivity Modulation of the Ley-Griffith TPAP Oxidation with N-Oxide Salts By: Moore, Peter W.; et al European Journal of Organic Chemistry (2016), 2016(20), 3401-3407.
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Scheme 31 (1 Reaction)

Steps: 1 Yield: 85%



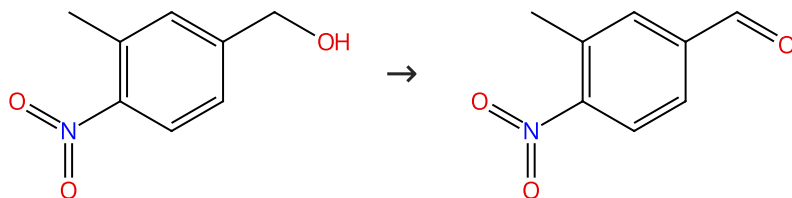
Suppliers (102)

Suppliers (89)

31-480-CAS-18797015 Steps: 1 Yield: 85% 1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt 1.2 Catalysts: 2204287-72-7; 0.5 h, rt Experimental Protocols	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis By: Moore, Peter W.; et al Chemistry - A European Journal (2018), 24(18), 4556-4561.
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Scheme 32 (1 Reaction)

Steps: 1 Yield: 84%



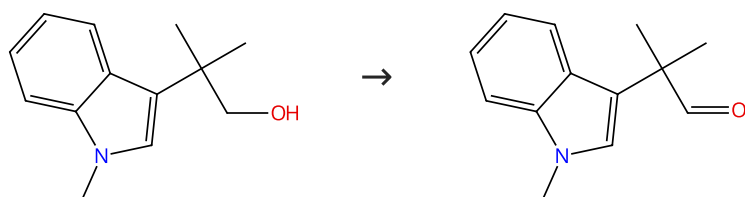
Suppliers (71)

Suppliers (65)

31-480-CAS-15777415 Steps: 1 Yield: 84% 1.1 Reagents: 4-Aza-1-azoniabicyclo[2.2.2]octane, 1-hydroxy-, tetraphenylborate(1-), compd. with 1,4-diazabicyclo[2.2.2]octane 1-oxide (1:1:1) Solvents: Acetonitrile; 10 min, rt 1.2 Catalysts: TPAP; 16 h, rt 1.3 Solvents: Water; rt Experimental Protocols	Selectivity Modulation of the Ley-Griffith TPAP Oxidation with N-Oxide Salts By: Moore, Peter W.; et al European Journal of Organic Chemistry (2016), 2016(20), 3401-3407.
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Scheme 33 (1 Reaction)

Steps: 1 Yield: 84%



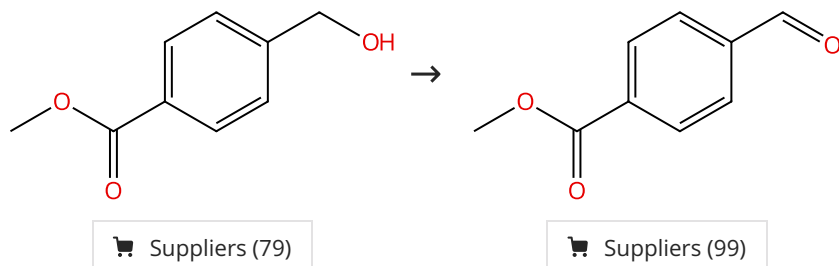
Suppliers (7)

Suppliers (8)

31-480-CAS-22932529	Steps: 1 Yield: 84%	Expeditious total synthesis of hemiassterlin through a convergent multicomponent strategy and its use in targeted cancer therapeutics
1.1 Reagents: N-methylmorpholine N-oxide Catalysts: TPAP Solvents: Dichloromethane; 20 h, rt		By: Charoenpattarapreeda, Jiraborrirak; et al
Experimental Protocols		Angewandte Chemie, International Edition (2020), 59(51), 23045-23050.

Scheme 34 (1 Reaction)

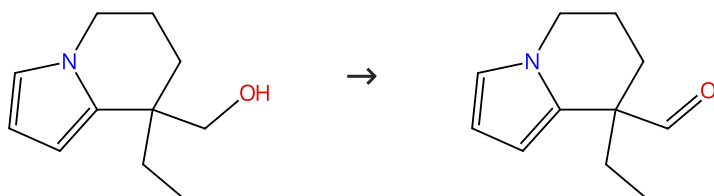
Steps: 1 Yield: 82%



31-480-CAS-15777413	Steps: 1 Yield: 82%	Selectivity Modulation of the Ley-Griffith TPAP Oxidation with N-Oxide Salts
1.1 Reagents: 4-Aza-1-azoniabicyclo[2.2.2]octane, 1-hydroxy-, tetraphenylborate(1-), compd. with 1,4-diazabicyclo[2.2.2]octane 1-oxide (1:1:1) Solvents: Acetonitrile; 10 min, rt		By: Moore, Peter W.; et al
1.2 Catalysts: TPAP; 16 h, rt		European Journal of Organic Chemistry (2016), 2016(20), 3401-3407.
1.3 Solvents: Water; rt		
Experimental Protocols		

Scheme 35 (1 Reaction)

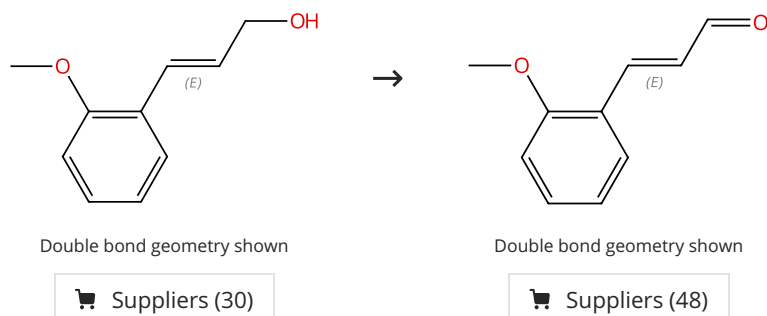
Steps: 1 Yield: 81%



31-480-CAS-17073752	Steps: 1 Yield: 81%	Formal Total Synthesis of (±)-Rhazinal: Evaluating the Radical Approach
1.1 Reagents: N-methylmorpholine N-oxide Catalysts: TPAP Solvents: Acetonitrile; 1 h, rt		By: Hildebrandt, Sven; et al
		Synthesis (2017), 49(11), 2584-2588.

Scheme 36 (1 Reaction)

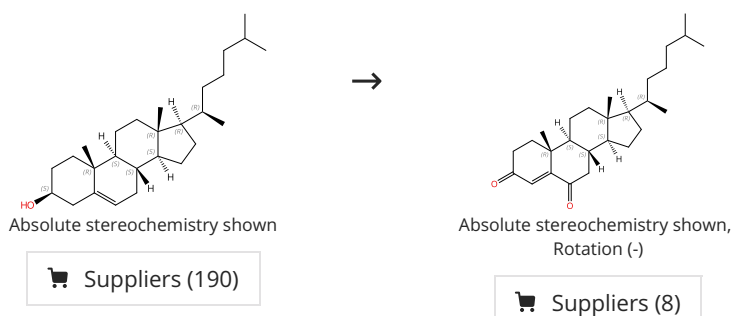
Steps: 1 Yield: 80%



31-480-CAS-15777416 Steps: 1 Yield: 80% 1.1 Reagents: 4-Aza-1-azoniabicyclo[2.2.2]octane, 1-hydroxy-, tetraphenylborate(1-), compd. with 1,4-diazabicyclo[2.2.2]octane 1-oxide (1:1:1) Solvents: Acetonitrile; 10 min, rt 1.2 Catalysts: TPAP; 16 h, rt 1.3 Solvents: Water; rt Experimental Protocols	Selectivity Modulation of the Ley-Griffith TPAP Oxidation with N-Oxide Salts By: Moore, Peter W.; et al European Journal of Organic Chemistry (2016), 2016(20), 3401-3407.
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Scheme 37 (1 Reaction)

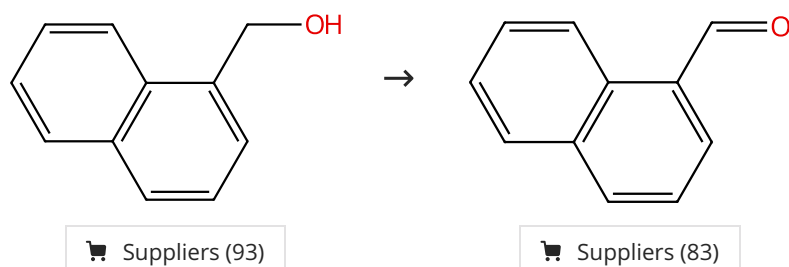
Steps: 1 Yield: 80%



31-480-CAS-19174895 Steps: 1 Yield: 80% 1.1 Reagents: N-methylmorpholine N-oxide Catalysts: TPAP; 90 min	Synthetic approaches to phomactins: Novel oxidation of homoallylic alcohols using tetra-n-propylammonium perruthenate By: Blackburn, Timothy J.; et al Tetrahedron (2018), 74(38), 5399-5407.
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Scheme 38 (1 Reaction)

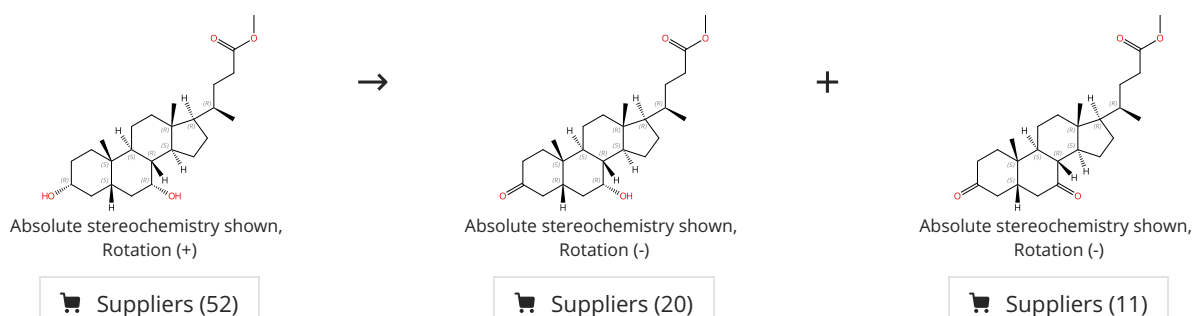
Steps: 1 Yield: 80%



31-480-CAS-18797010 Steps: 1 Yield: 80% 1.1 Reagents: N-methylmorpholine N-oxide Solvents: Dichloromethane; 1 h, rt 1.2 Catalysts: 2204287-72-7; 0.5 h, rt Experimental Protocols	ATP3 and MTP3: Easily Prepared Stable Perruthenate Salts for Oxidation Applications in Synthesis By: Moore, Peter W.; et al Chemistry - A European Journal (2018), 24(18), 4556-4561.
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Scheme 39 (1 Reaction)

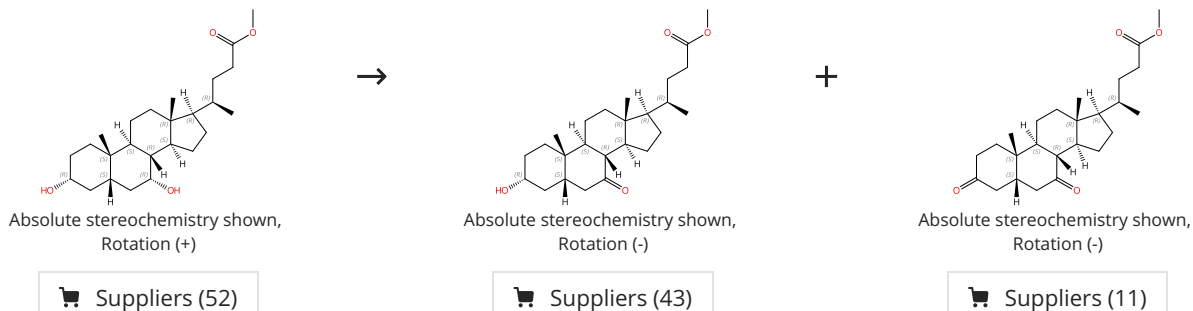
Steps: 1 Yield: 91%



31-614-CAS-34334477 Steps: 1 Yield: 91%	Selectivity of Oxidizing Agents toward Axial and Equatorial Hydroxyl Groups
1.1 Reagents: Sodium hypochlorite Catalysts: Tempo, Sodium bromide Solvents: Dichloromethane, Water; 0 °C; 0 °C → rt; 3 h, rt	By: Kaspar, Miroslav; et al Journal of Organic Chemistry (2022), 87(14), 9157-9170.
Experimental Protocols	

Scheme 40 (1 Reaction)

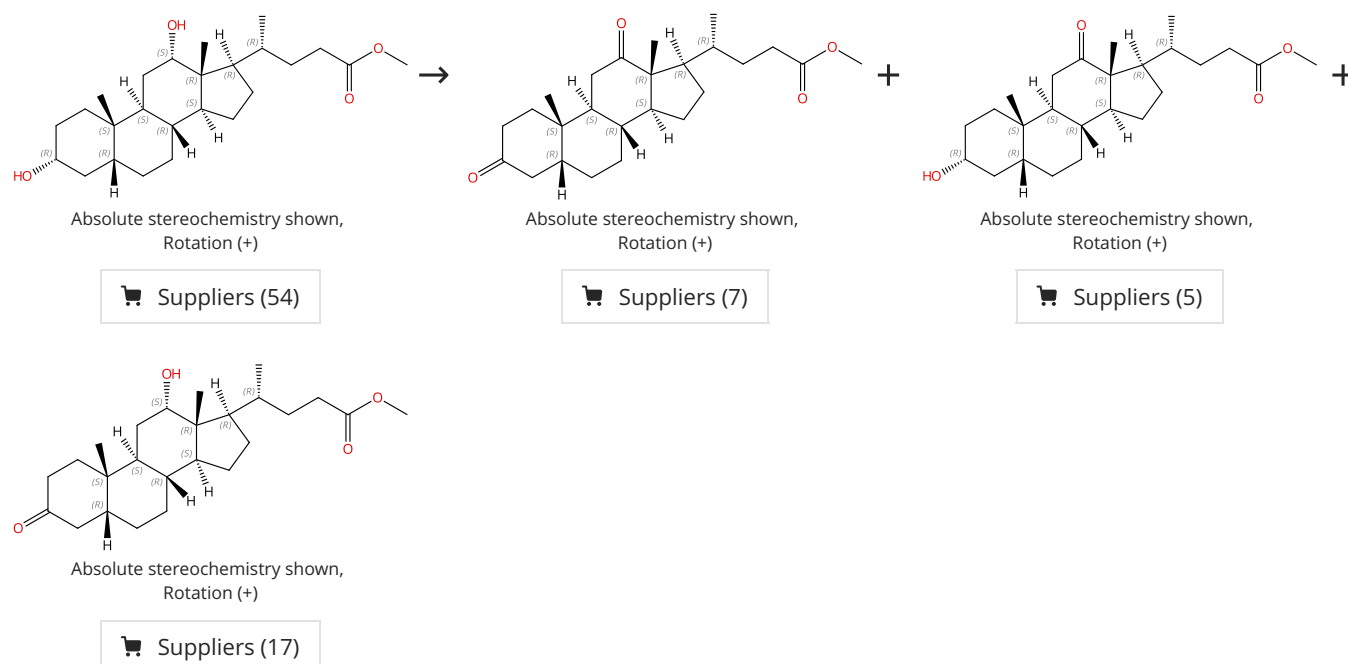
Steps: 1 Yield: 81%



31-614-CAS-34334481 Steps: 1 Yield: 81%	Selectivity of Oxidizing Agents toward Axial and Equatorial Hydroxyl Groups
1.1 Reagents: Pyridinium chlorochromate Solvents: Dichloromethane; 16 h, rt	By: Kaspar, Miroslav; et al Journal of Organic Chemistry (2022), 87(14), 9157-9170.
Experimental Protocols	

Scheme 41 (1 Reaction)

Steps: 1 Yield: 87%



31-614-CAS-34334479 Steps: 1 Yield: 87%	Selectivity of Oxidizing Agents toward Axial and Equatorial Hydroxyl Groups
1.1 Reagents: Iodobenzene diacetate Catalysts: Tempo Solvents: Dichloromethane; 4 h, 0 °C → rt	By: Kaspar, Miroslav; et al Journal of Organic Chemistry (2022), 87(14), 9157-9170.
Experimental Protocols	